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PRIMARY RESEARCH PAPER

Enhancing pond connectivity: impacts on common and rare species of macrophyte in an agricultural multi-pond system

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Abstract Previous studies have demonstrated differences between common and rare species in dispersal ability and reproductive strategies, leading to distinct responses to environmental and spatial processes. In this study, we collected data from 111 ponds within a multi-pond system, including environmental variables, land use, connectivity variables, and macrophyte species abundance. We aimed to investigate the influence of hydrological connectivity (watercourse dispersal) and geographical connectivity (overland dispersal) on the macrophyte species turnover in ponds, with a specific focus on common and rare species. Our findings emphasized the common macrophytes served as better indicators of species richness than rare species, suggesting that common species played a more significant role in shaping diversity patterns. Results showed that water flow, weighted by watercourse distance, accounted for 5.12% of the variance in species distribution, whereas geographical distance explained only 2.19%, indicating watercourse connectivity is more influential in structuring macrophyte communities than dispersal by wind or animals. For common species, water flow and environmental conditions were crucial for distribution, whereas rare species were influenced by a range of spatial variables. This study enhanced our understanding of ecological dynamics and conservation needs in freshwater habitats, emphasizing the importance of connectivity in supporting diverse aquatic species communities and ecological recovery.

Keywords Pond connectivity · Macrophyte common and rare species · Dispersal · Environmental filtering · AEM

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Introduction

Freshwater ecosystems support a diverse range of aquatic species (Faghihinia et al., 2021), with

macrophytes serving as primary producers and vital biological indicators (Søndergaard et al., 2010). Macrophytes play a pivotal role in structuring aquatic ecosystems by influencing nutrient cycling, sediment stability, and habitat complexity and contribute to ecosystem functions such as oxygen production, food web support, and water filtration (Madsen et al., 2001). Their ability to modulate light availability and create refuges for aquatic organisms further emphasizes their ecological importance (Engelhardt & Ritchie, 2001). As indicators of ecosystem health and sustainability (Thomaz et al., 2023), macrophytes can reflect changes in nutrient levels, water clarity, pollution levels, and other ecological factors through shifts in species composition, abundance, and growth patterns (Sun et al., 2019).

The distribution of macrophytes in lake networks is often constrained by local-scale environmental filtering (Heino et al., 2007; Alahuhta & Heino 2013; Sun et al., 2022), as well as regional-scale spatial processes governing dispersal (Shmida & Wilson, 1985; Bouvier et al., 2009). Extensive research has demonstrated the importance of different spatial processes for maintaining regional biodiversity in aquatic species (Naiman et al., 1993; Akasaka & Takamura, 2011; Chapman et al., 2019). Dispersal modes and habitat connectivity were found to be critical determinants of macrophyte species composition in freshwater ecosystems (Chambers et al., 2008).

Macrophyte dispersal modes are typically categorized into three primary mechanisms: aquatic active dispersal via water flow, aerial active dispersal by wind, and aerial passive dispersal through aquatic insects or adult flight (Poff et al., 2006; Schmidt-Kloiber & Hering, 2015). The dispersal capacity of macrophyte species can be viewed as a complex trait influenced by these various mechanisms, which determine their ability to colonize new environments and maintain genetic diversity across fragmented habitats (Bornette & Puijalon, 2011). (i) Water transports aquatic propagules through streams and the floodplain network (Boedeltje et al., 2004; Merritt & Wohl, 2006; Chen et al., 2007). This unidirectional dispersal explains why downstream habitats exhibit higher biodiversity (Gornall et al., 1998; Liu et al., 2006). Dispersal distance is promoted by increasing river discharge, such as floods (Nilsson et al., 1991), and the duration of seed buoyancy (Boedeltje et al., 2004). Moreover, the floating ability of seeds also plays a vital role in the dispersal process for aquatic vegetation (Schmiedel & Tackenberg, 2013). (ii) Wind is a key mechanism for the aerial dispersal of macrophyte seeds and spores, enabling colonization of isolated water bodies, particularly in areas with limited water connectivity. The small, lightweight propagules can travel long distances, allowing macrophytes to expand their range and promote genetic exchange across fragmented habitats (Soons et al., 2004). This mechanism supports biodiversity by facilitating recolonization after disturbances and connecting isolated populations (Nathan & Muller-Landau 2000). Wind dispersal helps maintain ecosystem resilience and stability, promoting diverse and adaptable macrophyte communities in both connected and isolated ecosystems (Moi et al., 2021). (iii) Water-dwelling animals, especially water birds, also play a crucial role in dispersing aquatic plants (Green et al., 2008). Propagules can be transported to new sites by attaching to birds' feet or via ingestion followed by excretion. This mode of dispersal depends on factors such as bird density, feeding habits, and migratory behavior (Viana et al., 2016). The movement patterns of these animals, particularly their migratory behaviors, enable macrophytes to colonize distant ecosystems, enhancing genetic exchange and species diversity across regions (Viana et al., 2013).

Habitat connectivity also influences the distribution patterns of macrophyte assemblage (Hughes et al., 2009). This connectivity can be measured in terms of geographical or hydrological connectivity of the ditch network among waterbodies. Assessing potential lake connectivity is essential for understanding macrophyte dispersal and colonization (Guimaraes et al., 2014). Different landscape indices and distance metrics have been utilized to evaluate the spatial connectivity between waterbodies (Morris, 2012). Geographical connectivity, for instance, is used to estimate the potential impact of wind or water birds in determining the mobility of aquatic plants in the landscape (Hartvigsen & Kennedy, 1993; Ganio et al., 2005; Miyazono et al., 2010). In some cases, overland distance between lakes best explained gene flow for some species, while for others, both watercourse and overland distance were highly correlated (Alp et al., 2012). However, the aerial dispersal may also play a significant role in dispersal following the river

network in a hydrologically connected system rather than a disconnected one. Hydrological connectivity, measured as the continuous watercourse length between two lakes, is a major pathway for water-mediated dispersal of plants across the landscape (Johansson et al., 1996; Ward et al., 2002). Fluvial corridors input a large number of plant seeds via drainage from floodplains and provide food and habitat for water birds, increasing the probability of seed carriage (Beier & Noss, 1998; Ishiyama et al., 2014). Flood events disperse plant propagules to floodplain lakes and influence the species sorting process (Thomaz et al., 2007). Conversely, the loss of hydrological connectivity, caused by damming for example, disrupts the movement of propagules to downstream lakes by altering the hydrological regime (Jansson et al., 2000; Merritt & Wohl, 2006). For species that rely heavily on hydrological connectivity between habitats, dispersal via wind and animal migration may be less critical (De Bie et al., 2012), as aquatic species can rapidly colonize habitats with high levels of hydrological connectivity (Capers et al., 2010).

In ecological communities, common species are typically those that are widely distributed and abundant across the landscape, while rare species are characterized by limited distribution, low abundance, or specialized habitat requirements (Gaston, 1994). In macrophyte communities, a small proportion of common species and a large proportion of rare species are responsible for the specific pattern of rarity (Flather et al., 2007). Common macrophyte species, owing to their widespread distribution and abundance, serve as reliable indicators of ecosystem health and provide early warning signs of environmental degradation (Lennon et al., 2004; Landi & Chiarucci, 2014). Rare macrophytes, on the other hand, are often highly sensitive to specific environmental conditions, making them valuable indicators of localized or specialized habitats (Gaston, 1994). Previous studies have shown that common and rare species differ in dispersal ability (Siqueira et al., 2012), reproductive strategies (Kunin & Gaston, 1993; Kunin & Shmida, 1997), and environmental niche (Cornwell & Ackerly, 2010). These differences result in varying responses to environmental conditions and spatial processes, with common species often being more resilient to habitat changes and disturbances, while rare species may be more vulnerable to environmental fragmentation and connectivity issues (Gaston & Fuller, 2008). Most studies agreed that the determinants of commonness and rarity were related to niche differences, such as environmental niche and competitive ability (Magurran & Henderson, 2003; Cornwell & Ackerly, 2010), coupled with processes that govern regional-scale dispersal, such as species dispersal ability (Rabinowitz & Rapp, 1985; Kunin & Gaston, 1993), hydrological isolation, or flooding (Shmida & Wilson, 1985; Bouvier et al., 2009; Sun et al., 2018). However, dispersal mechanisms alone cannot fully explain species distributions; abiotic factors, such as water availability, nutrient levels, and habitat suitability, also play crucial roles (Grime, 2001). For instance, species with strong dispersal abilities may still be rare if the abiotic conditions in the area do not support their establishment and growth (Catella & Abbott, 2023).

Despite the recognized importance of both biotic and abiotic factors, few studies have examined how habitat connectivity and abiotic conditions jointly affect common and rare macrophyte species, particularly rare species, which are often omitted in community-level analyses (Cottenie & De Meester, 2003). Understanding the connectivity–dispersal relationship, alongside the role of local abiotic conditions, is essential for explaining the distribution patterns of macrophyte communities. These patterns depend not only on the connectivity of the landscape and species-specific dispersal strategies but also on the suitability of the environmental conditions in which they disperse. In this work, we briefly discussed the dispersal processes of common and rare macrophyte species occurring along the watercourse or overland. Common species typically benefit from broader dispersal pathways, exhibiting greater tolerance to diverse environmental conditions and occupy wider ecological niches (Lennon et al., 2004). From this, we derive our first hypothesis: (1) Common species contribute more significantly to overall macrophyte species richness than rare species in the multi-pond system. Environmental factors like nutrient levels, water transparency, and habitat structure are expected to influence common species more, as they respond to broad ecological gradients. This leads to our second hypothesis: (2) Environmental variables more effectively explain the species composition of common species than that of rare

species. The third hypothesis—(3) Watercourse-mediated dispersal processes are closely associated with both common and rare species—builds on the premise that hydrological connectivity between ponds plays a key role in shaping their community structure. This study provided novel insights into how rare and common macrophyte species differ in niche occupation and propagation strategies, demonstrating their distinct roles in maintaining wetland ecosystem stability.

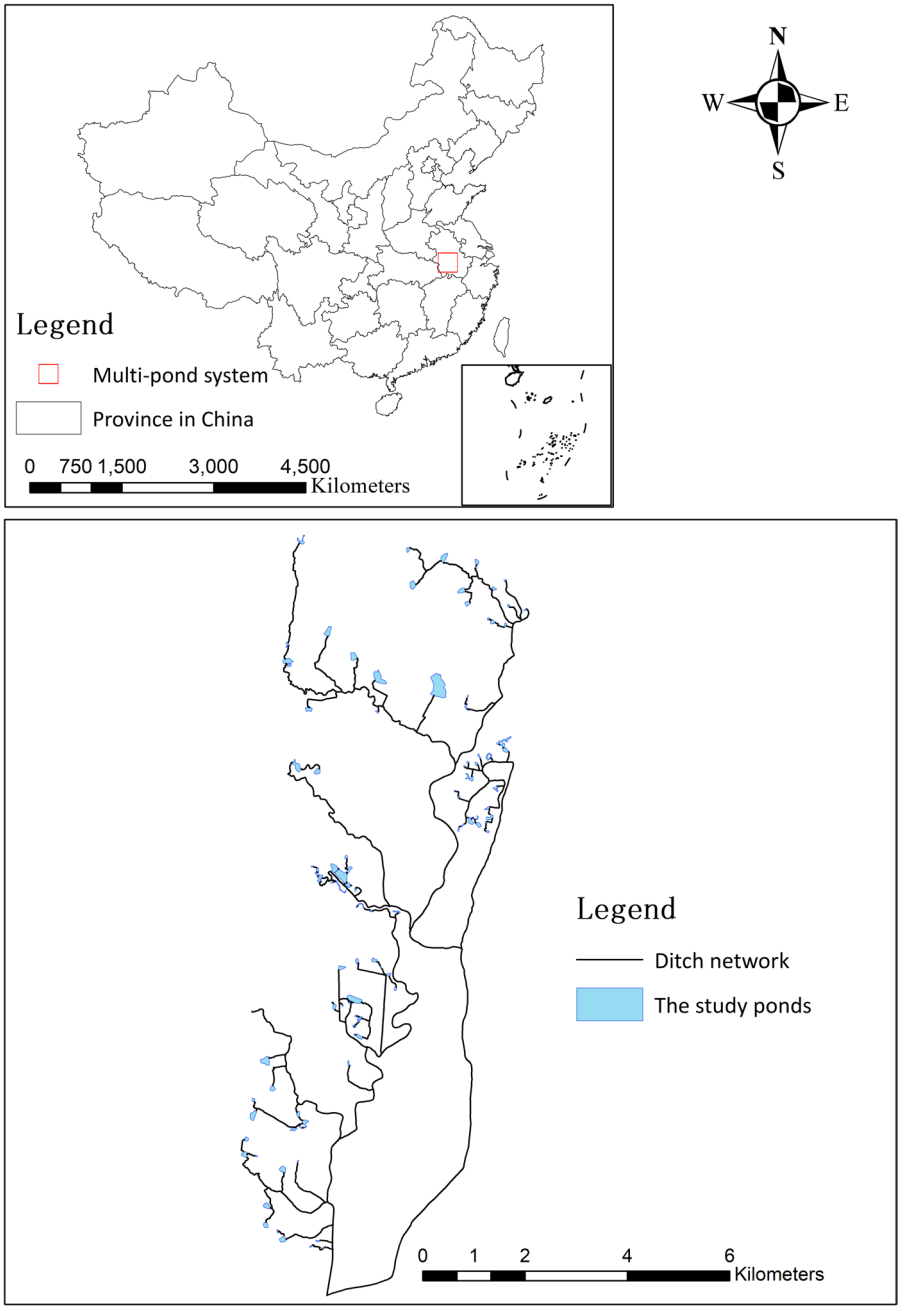


Fig. 1 The geographic location of the study ponds in the agricultural multi-pond system

Materials and methods

Study areas

A comprehensive survey was conducted on 111 ponds within a multi-pond system situated in a small tributary of the Yangtze River, encompassing the catchment of Caizi Lake in China (Fig. 1). The multi-pond system serves as an exceptional example of an artificial wetland within an agricultural landscape, connecting hundreds of artificial ponds through ditches, enabling the transfer of energy, materials, and aquatic species (Yin & Shan, 2001; Wang et al., 2023). The study multi-pond system is subjected to the subtropical monsoon climate, leading to a significant amount of precipitation during the summer. The study area is relatively flat, with an elevation range of 1–2 m, and the ponds within the system are shallow, with depths ranging from 0.5–2.5 m. The main land use types in the study area are cultivated land and urban areas. All ponds in this system are interconnected via the ditches and provide suitable habitats for waterbirds and other aquatic communities. Additionally, the area is prone to frequent flooding, mainly due to its function as a flood discharge zone for the Yangzi River.

The scattered ponds in the multi-pond system serve a crucial function in supporting agricultural irrigation, aquaculture, and human activities such as washing for local inhabitants (Needham, 1984; Yin et al., 1993). Typically, each pond is responsible for irrigating approximately 1–2 hectares of nearby cultivated land. Furthermore, the multi-pond system functions as an agricultural runoff recipient, intercepting nitrogen and phosphorus inputs from surrounding farmlands. By temporarily retaining and diluting these nutrients, the pond mitigates non-point source pollution and alleviates nutrient loading in downstream water bodies, thereby contributing to the stability of freshwater ecosystems (Yin & Shan, 2001). In this system, water can flow from high-altitude ponds to lower-altitude ones without passing through the nearby farmland. Despite its ecological significance, the multi-pond system has suffered severe degradation due to human disturbances such as agricultural intensification and ditch hardening (Li et al., 2020).

Extraction of the study ponds and ditches

In the multi-pond system, ponds interconnected by ditches are dispersed throughout the cultivated land (Figure S1). To extract all ponds and ditches in the study area, high-resolution satellite images (with a resolution up to 0.256 m) were downloaded from LocaSpaceViewer4 based on the Landsat 8 image (08/11/2019). The shapefiles including the ponds and watercourses layers were created and edited using ArcGIS. The extracted data were verified in the fieldwork, resulting in an accuracy of 93.56%. A total of 111 ponds were selected in the study area with interconnected ditches (Fig. 1). Pond selection was based on two main criteria: (i) hydrological connectivity—each pond had to be effectively hydrological connected with other ponds via ditches, enabling water-driven macrophyte dispersal; (ii) spatial dispersion—ponds were distributed across the landscape to represent environmental gradients and support natural species movement within the interconnected system. To ensure that the selected ponds were representative of the broader multi-pond system, selection prioritized variation in size, shape, and degree of connectivity, thereby capturing the landscape's heterogeneity. This physical diversity reflects the ecological variability across the study area and provides a robust basis for analyzing macrophyte species distribution. Additionally, spatial dispersion was considered to ensure that the selected ponds were well-distributed across the study region, capturing broader-scale variation and accounting for alternative dispersal mechanisms, such as wind and animal vectors, that may influence species distribution beyond water-mediated transport.

To assess land use patterns in the study area, we used high-resolution satellite imagery from two sources: Landsat 8 data acquired from the USGS platform and Google Earth imagery obtained via LocaSpaceViewer4. These images were used to quantify the extent of urban and agricultural land use within a 200-m buffer zone surrounding each pond, which served as a useful proxy for assessing the level of anthropogenic disturbance. We utilized ArcGIS v10.3 to extract and calculate land use metrics for each pond. Specifically, we focused on two categories: urban land use, including suburban and continuously build-up areas, and agricultural land use, comprising cultivated land, arable cereals,

and non-rotational arable crops. This information was essential for evaluating the extent of human activities and their potential impacts on macrophyte communities in the study ponds.

Water chemistry sampling

In September 2020, we investigated the water chemistry and macrophyte communities of 111 study ponds in the agricultural multi-pond system. Water quality parameters, such as temperature (T), the concentration of dissolved oxygen (DO), salinity (SAL), the electrical conductivity (Cond), oxidation–reduction potential (ORPmv), and pH, were directly measured from the shoreline of the pond using the YSI Multi-parameter Water Quality Analyzer (Model YSI 6600, USA). To analyze water chemistry, samples were stored in clean and sterile bottles, kept refrigerated, and transported to the laboratory for analysis. Total nitrogen (TN) was measured using the modified Kjeldahl method (Sáez-Plaza et al., 2013), while total phosphorus (TP) was determined using the molybdenum blue method (Murphy & Riley, 1962). Chlorophyll *a* (Chl-*a*) concentrations were analyzed spectrophotometrically following the method described by Jeffrey & Humphrey (1975). In our study, Secchi disk measurements from the shoreline of small ponds helped assess water transparency (Tilzer, 1988). The area and depth of each pond were calculated based on the extracted pond shapefile in ArcGIS. Elevation of each pond was obtained from the Shuttle Radar Topography Mission (SRTM), with a pixel size of 30-m. The statistical analyses of each environmental variable are presented in Table 1.

Macrophyte sampling

In each of the 111 study ponds, macrophytes were sampled from the shallow littoral zones using 1 × 1 m quadrats. Five to seven quadrats were randomly distributed along the shoreline for adequate representation. Submerged plants were retrieved with a hook, and for plants in the quadrats, macrophytes were collected by wading and using large tongs. In the limnetic region, three sites were sampled: two quadrats along opposite shorelines and one in the central zone (Sun et al., 2022). The stainless steel aquatic plant sampler (DXCC-3850) was used for submerged plants in the water. Shoreline and riparian plants, including those in the water level fluctuation zone, were also sampled with 1 × 1 m quadrats. All plants were identified to species in the field, with voucher specimens and photographs collected for difficult-to-identify species (Cook, 1990; Wang et al., 2021). A total of 52 macrophyte species were recorded across all ponds in the study (Table S1). The total species richness was subsequently calculated for each pond. It should be noted that submerged plants were scarcely observed in the study area, likely due to non-point source pollution such as herbicides and pesticides that have led to the destruction of original vegetation in many ponds.

Defining macrophyte common and rare species

The classification of common and rare macrophyte species followed established ecological criteria (Gaston, 1994; Magurran, 2004). Rare species were identified according to the lower quartile of species abundance, while common species corresponded to the upper quartile. However, distinguishing between common and rare species can be challenging, as their distribution may vary across different habitats and ecosystems.

Two approaches of the infection point approach and Gaston's quartile were used to distinguish the common and rare macrophyte species in this study (Gaston, 1994; Siqueira et al., 2012; Samways & Pryke, 2016). First, we calculated a rank abundance curve for the macrophyte dataset and identified the inflection point, where the curve's curvature changed. Species to the left of the inflection point were classified as common, while those to the right were considered rare (Siqueira et al., 2012). Although the precise location of the division in relative abundance can vary, studies have shown that this does not significantly affect conclusions (Magurran, 2004). In this study, we followed Siqueira et al. (2012) and defined the top 25% of abundant species as common, and the remaining three quartiles (25%, 50%, and 75%) as rare (Table 2). Using these definitions, the corresponding matrix of the common species and rare species for each pond was calculated.

Table 1 Statistical analyses of all explanatory variables in the study ponds

Datasets	Variables	Abbreviation	Unite	Definition	Source	Range	Mean ± SD
Environment	Mean annual temperature	T	℃	Mean annual temperature of air at 2 m for each pond from 2010 to 2019	GEE-ERA5-Land Monthly Aggregated	17.239–17.258	17.242 ± 0.0075
	Dissolved oxygen	DO	mg/L	Dissolved oxygen concentration of pond	YSI	1.14–12.43	6.41 ± 2.74
	Conductivity	Cond	VS/cm	Conductivity concentration of pond water	YSI	68.7–370	159.48 ± 48.23
	Salt	SAL	ppt	Salt value of pond water	YSI	0.03–0.18	0.078 ± 0.025
	pH	pH	-	pH value of pond water	YSI	4.46–9.38	7.75 ± 0.88
	Oxidation–Reduction Potential	ORPmv	mV	ORP is commonly used to assess water quality	YSI	10–231.1	79.86 ± 31.95
	TN	TN	ug/ml	Total nitrogen concentration of pond water	Lab measure	0.041–5.32	1.31 ± 0.91
	TP	TP	ug/ml	Total phosphorus concentration of pond water	Lab measure	0.017–0.92	0.17 ± 0.12
	Chla	Chla	ug/L	Chlorophyll concentration of pond water	Lab measure	3.21–251.08	69.02 ± 53.25
	Elevation	ELE	m	Pond elevation	GIS	10.73–36.04	17.99 ± 4.57
	Pond area	AREA	m^2	Area of pond	GIS	242–97,947	6467 ± 10,798
	Water transparency	WT	cm	Transparency of pond water	Secchi disk	15–80	35.5 ± 13.4
	Water depth	Depth	cm	Mean depth of pond	GIS	20–450	110 ± 86
Land use	Building proportion	Building%	%	The extent of urban land use within a 200-m buffer surrounding each pond	Landsat 8 & LocaSpaceViewer4	0–0.26	0.039 ± 0.049
	Farmland proportion	Farmland%	%	The extent of agricultural land use within a 200-m buffer surrounding each pond	Landsat 8 & LocaSpaceViewer4	0.11–0.91	0.52 ± 0.19
	The number of buildings	Building ID	–	The number of buildings within a 200-m buffer zone surrounding each pond	Landsat 8 & LocaSpaceViewer4	0–110.1	19.7 ± 22.1

Table 1 (continued)

Datasets	Variables	Abbreviation	Unite	Definition	Source	Range	Mean ± SD
Hydrology connectivity	Watercourse distance-based MEM	WCDB-MEM	–	The non-directional dispersal via watercourse distance	GIS & R	–	–
	0/1 AEM	0/1-AEM	–	The 0/1 based AEM emphasizes flow direction	GIS & R	–	–
	Watercourse distance-based AEM	WCDB-AEM	–	The weighted AEM emphasizes both flow direction and watercourse distance	GIS & R	–	–
	Pond runoff inflow	Inflow	m^3/day	Q_{runoff} is mainly influenced by rainfall and runoff (Chow, 1959). Q_{runoff} = Runoff Coefficient × Precipitation (m/d) × Pon d Area (m^2)	GEE-ERA5-Land Monthly Aggregated	9.98–4040	267.9 ± 445.7
	Pond Evaporation Outflow	Outflow	m^3/day	Q_{evap} mainly influenced by the evaporation rate and the pond area (Brutsaert, 1982). Q_{evap} = the evaporation rate (m/d) × pond area (m^2)	GEE-ERA5-Land Monthly Aggregated	10.05–3874	259.7 ± 430.9
	Pond Water Retention time	WRT	Day	$T_{residence}$ refers to the time required for pond water to be completely replaced (Chapra, 2008). $T_{residence}$ = Pond water volume (m^3)/(inflow(m^3/d)-outflow(m^3/d))	GEE-ERA5-Land Monthly Aggregated	11.75–1146	136.3 ± 190.8
Landscape connectivity	Moran's Eigenvector Maps	MEM	–	The non-directional dispersal through overland distance	ArcGIS	–	–
	Probability of Connectivity	PC	–	The likelihood that a species can move between two ponds within a landscape	ArcGIS	0.067–26.7	1.75 ± 2.95
	Integral Index of Connectivity	IIC	–	The degree of connectivity between ponds based on the overland distance	ArcGIS	0.088–26.46	1.78 ± 2.95

Table 2 The rank abundance of all macrophyte species in the study ponds using the 'rankabundance' function in 'BiodiversityR' package in R

Species name	Rank	Abundance	Plower	Pupper	Accumfreq	Logabun	Rankfreq
Alternanthera philoxeroides (Mart.) Griseb	1	406	21.1	26.7	23.9	2.6	1.6
Lemna minor L	2	148	6.5	11	32.6	2.2	3.2
Paspalum paspaloides (Michx.) Scribn	3	128	5.7	9.3	40.1	2.1	4.8
Phragmites australis (Cav.) Trin. ex Steud	4	126	5.6	9.2	47.5	2.1	6.5
Typha orientalis C. Presl	5	94	3.8	7.3	53.1	2	8.1
Marsilea quadrifolia L	6	74	3	5.7	57.4	1.9	9.7
Bidens pilosa L	7	63	2.9	4.5	61.1	1.8	11.3
Polygonum hydropiper L	8	61	2.5	4.7	64.7	1.8	12.9
Polygonum orientale L	9	43	1.7	3.4	67.2	1.6	14.5
Trapa natans L	10	38	1	3.5	71.9	1.6	17.7
Cyperus iria L	11	37	1.6	2.8	74.1	1.6	19.4
Hydrocharis dubia (Blume) Backer	12	36	1.2	3	76.2	1.6	21
Setaria viridis (L.) P. Beauv	13	33	1.1	2.8	78.1	1.5	22.6
Ipomoea aquatica Forssk	14	30	1	2.5	79.9	1.5	24.2
Echinochloa crusgalli (L.) P. Beauv	15	27	1	2.2	81.5	1.4	25.8
Salvinia natans (L.) All	16	24	0.5	2.4	82.9	1.4	27.4
Leersia parviflora Desv	17	18	0.1	2	83.9	1.3	29
Digitaria sanguinalis (L.) Scop	18	17	0.5	1.5	84.9	1.2	30.6
Eichhornia crassipes (Mart.) Solms	19	16	0.4	1.5	85.9	1.2	32.3
Polygonum coriarium Grig	20	15	0	1.7	86.8	1.2	33.9
Acorus calamus L	21	15	0.2	1.6	87.6	1.2	35.5
Conyza bonariensis (L.) Cronquist	22	15	0.2	1.5	88.5	1.2	37.1
Imperata cylindrica (L.) P. Beauv	23	14	0.3	1.3	89.4	1.1	38.7
Nelumbo nucifera Gaertn	24	14	0.1	1.6	90.2	1.1	40.3
Zizania latifolia (Griseb.) Turcz. ex Stapf	25	11	0.3	1	90.8	1	41.9
Eclipta prostrata (L.) L	26	10	0.2	1	91.4	1	43.5
Eleusine indica (L.) Gaertn	27	9	0.2	0.9	91.9	1	45.2
Elymus dahuricus Turcz	28	9	0.2	0.9	92.5	1	46.8
Fimbristylis miliacea (L.) Vahl	29	9	0.2	0.9	93	1	48.4
Myriophyllum verticillatum L	30	8	− 0.2	1.1	93.5	0.9	50
Ceratophyllum demersum L	31	8	0	1	93.9	0.9	51.6
Utricularia Vulgaris L	32	8	− 0.1	1.1	94.4	0.9	53.2
Juncus effusus L	33	7	0.1	0.7	94.8	0.8	54.8
Polygonum hydropiper L	34	7	0.1	0.7	95.2	0.8	56.5
Leptochloa chinensis (L.) Nees	35	7	0.1	0.7	95.6	0.8	58.1
Ludwigia prostrata Roxb	36	6	0.1	0.6	96	0.8	59.7
Cynodon dactylon (L.) Persoon	37	6	0.1	0.6	96.4	0.8	61.3
Alternanthera sessilis (L.) R. Br. ex DC	38	6	0.1	0.6	96.7	0.8	62.9
Rumex acetosa L	39	6	− 0.1	0.8	97.1	0.8	64.5
Pennisetum alopecuroides (L.) Spreng	40	5	0	0.6	97.4	0.7	66.1
Miscanthus floridulus (Labill.) Warburg ex K. Schumann	41	5	0	0.6	97.6	0.7	67.7
Nymphoides peltate (S. G. Gmel.) Kuntze	42	5	− 0.3	0.9	97.9	0.7	69.4
Sporobolus fertilis (Steud.) Clayton	43	4	0	0.5	98.2	0.6	71
Carex tenuispicula Tang ex S. Yun Liang	44	4	− 0.2	0.7	98.4	0.6	72.6
Monochoria korsakowii Regel & Maack	45	3	0	0.4	98.6	0.5	74.2

Table 2 (continued)

Species name	Rank	Abundance	Plower	Pupper	Accumfreq	Logabun	Rankfreq
Ceratopteris pteridoides (Hook.) Hieron	46	2	0	0.3	98.7	0.3	75.8
Kalimeris indica (L.) Sch.-Bip	47	2	0	0.3	98.8	0.3	77.4
Euryale ferox Salisb. ex K. D. Koenig & Sims	48	2	– 0.1	0.3	98.9	0.3	79
Paspalum thunbergii Kunth ex Steud	49	2	0	0.3	99.1	0.3	80.6
Rottboellia exaltata L	50	2	0	0.3	99.2	0.3	82.3
Potamogeton maackianus A. Benn	51	2	0	0.3	99.3	0.3	83.9
Najas minor All	52	2	0	0.3	99.4	0.3	85.5
Hemerocallis fulva L	53	2	0	0.3	99.5	0.3	87.1
Zephyranthes candida (Lindl.) Herb	54	1	– 0.1	0.2	99.6	0	88.7
Miscanthus sacchariflorus (Maxim.) Benth. & Hook. f. ex Franch	55	1	– 0.1	0.2	99.6	0	90.3
Utricularia Aurea Lour	56	1	– 0.1	0.2	99.7	0	91.9
Hydrilla verticillate (L. f.) Royle	57	1	-0.1	0.2	99.8	0	93.5
Chara fragilis Desv	58	1	– 0.1	0.2	99.8	0	95.2
Marsilea quadrifolia L	59	1	– 0.1	0.2	99.9	0	96.8
Equisetum arvense L	60	1	– 0.1	0.2	99.9	0	98.4
Cyperus rotundus L	61	1	– 0.1	0.2	100	0	100

Pond connectivity assessment

Spatial eigenfunction analyses were used to assess the spatial structure of the hydrological connected ponds in the multi-pond system. Spatial eigenfunction analyses captured multi-scale spatial patterns, enabling the identification of ecological processes operating at different spatial extents (Griffith & Peres-Neto, 2006). The methods assumed that spatial processes are continuous and can be represented by distance matrices, from which eigenvectors were derived to model spatial structures (Legendre & Legendre, 2012). Among these methods, Moran's eigenvector maps were widely applied as they maximize spatial autocorrelation using Moran's I and effectively captured both broad- and fine-scale spatial patterns, making them suitable for irregular sampling designs and reducing the risk of autocorrelation bias in ecological studies (Dray et al., 2006). However, the methods might suffer from collinearity between eigenvectors and environmental variables, potentially complicating result interpretation (Peres-Neto & Legendre, 2010). Additionally, the use of inappropriate spatial distance matrices or retention of an excessive number of eigenvectors may lead to biased results and overfitting (Blanchet et al., 2008). Spatial eigenfunction analyses involve a group of variables that induce clustering in the distribution of response variable values. Spatial patterns provide insights into the relative contributions of dispersal limitation and other spatial processes in shaping community composition, independent of the effects of environmental filtering. Euclidean distance represents the straight-line distance separating two focal ponds in a geographic context, while watercourse distance denotes the distance between the ponds as measured along the stream network. The concept of connectivity captures the extent to which the surrounding landscape facilitates or hinders movement between distinct pond habitats (Fig. 2).

Moran's eigenvector maps (MEM) were employed to model the spatial structure of the study ponds along the Euclidean distance (Fig. 2a) and watercourse distance (Fig. 2b). The distance matrix was generated by calculating the length of the river watercourse between all pairs of study ponds using ArcGIS (Borcard et al., 2011; Dray et al., 2006). To quantify the exchange of aquatic species among the study ponds, we developed a weighting matrix (Borcard et al., 2011) that incorporated both the connectivity and distance matrices (Fig. 2b). The weighting matrix was calculated using the function: $\text{weight} = 1\text{-}(d/d_{max})^3$, where d represents the length of watercourse between

the connected sampling sites, and d_{max} refers to the largest length of watercourse in the dataset.

Asymmetric eigenvector map (AEM) is a type of spatial eigenfunction that models directional spatial processes (Blanchet et al., 2008). It is specifically designed to capture the spatial structure of species distribution under directional forcing, such as water flow and river network environments. AEM eigenvectors do not directly measure spatial processes, but they are a special case of MEM and cannot be detrended when response variables are included (Borcard et al., 2011). The AEM model generates a set of spatial eigenvectors based on the directional connection diagram among the sampling ponds, with the links of the AEM weighted to represent the difficulty of dispersal among site pairs based on watercourse distance or flow accuracy (Blanchet et al., 2011). In contrast to the MEM model, detrending of the species data is not necessary in AEM, as the trend is an essential component of the directional spatial patterns (Blanchet et al., 2011). In AEM model, the flow direction is considered in constructing the connection diagram, and it is important to maintain the trend of the data in the AEM analyses to accurately reflect the relationship between spatial structure and community composition. To generate the Sites(ponds)-by-Edges(watercourse) matrix, we used a schematic map of the river network in the multi-pond system (Fig. 2c). For n sampling sites, there are n-1 spatial variables created, but only AEMs that model positive spatial correlation can be used in the analysis. The weights could also be assigned to the Edges of the matrix based on the watercourse distance (Fig. 2d), which reflect the difficulty of exchange between pairwise sampling sites. The weighting function was the same as the one used in MEM eigenvectors.

The rainfall regime was a key factor influencing both nutrient concentrations and hydrological connectivity within pond systems. Surface runoff, driven by precipitation, played a crucial role in shaping macrophyte communities by affecting water quality and nutrient availability. In our study, we incorporated several important hydrological variables into our dataset, including pond runoff inflow, pond evaporation outflow, and pond water retention time (Table 1). Additionally, we calculated the Integral Index of Connectivity (IIC) and the Probability of Connectivity (PC) based on pond locations to assess how spatial connectivity and overall network structure influence macrophyte dispersal. These metrics were included in the landscape connectivity dataset, providing a comprehensive analysis of connectivity effects on aquatic plant dynamics.

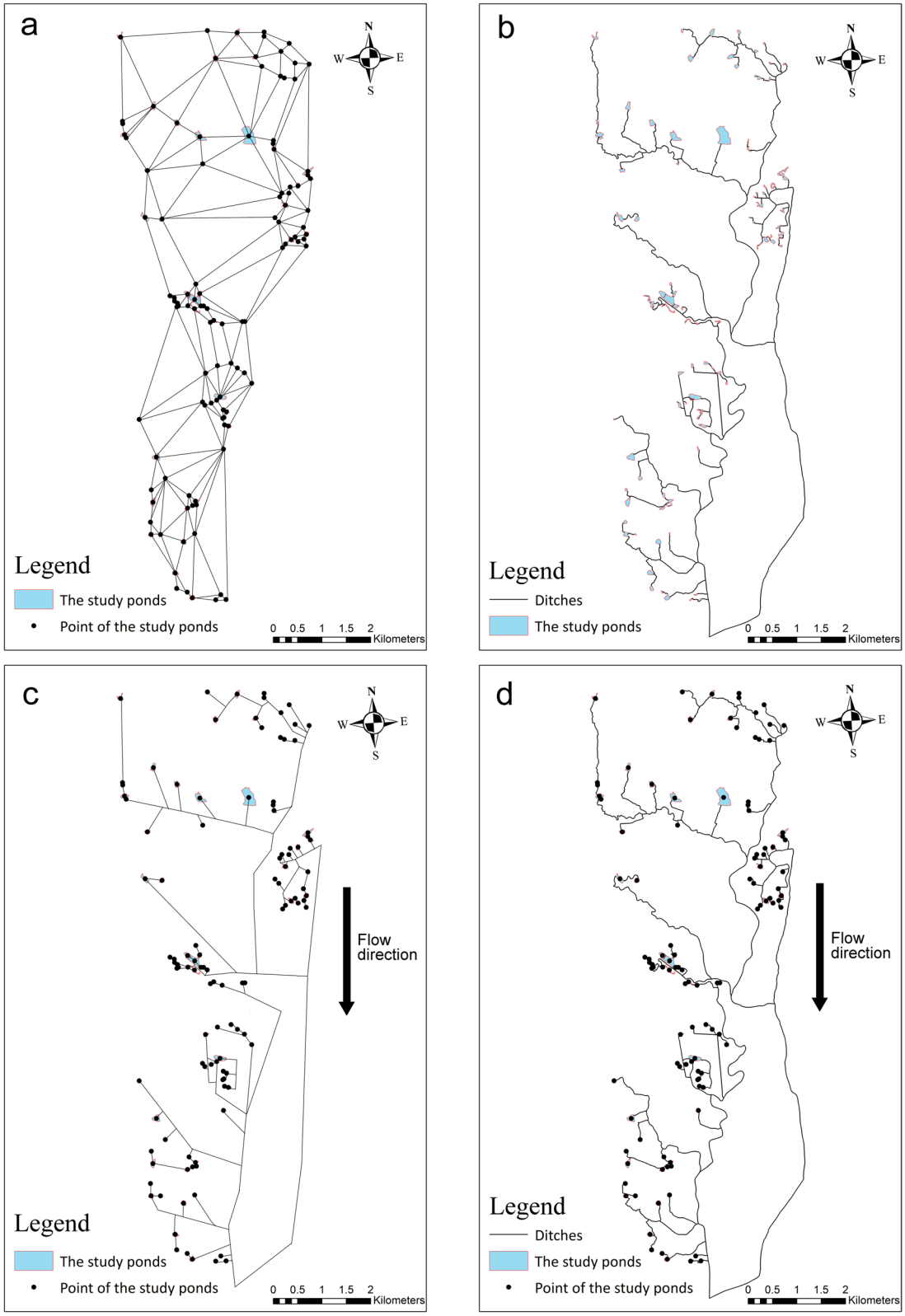
Statistical analyses

The macrophyte abundance species were ranked from most common to rarest according to the rank abundance curve (Brown, 1984; White et al., 2022) to produce a sequence of species richness patterns. At each step, one species was added from the most common species (i.e., sub-assemblage), and Pearson's correlation was calculated between the sub-assemblage richness pattern and the full assemblage (Lennon et al., 2004). This process was repeated for the rarest to most common species (Van Schalkwyk et al., 2019). The resulting sequence of correlations was plotted against the number of macrophyte species in the sub-assemblage to compare the contributions of common and rare species to overall patterns of species richness. Studying the relative contributions of common and rare species to species richness could help us understand the diversity patterns and ecological dynamics of macrophyte communities.

In this study, we aimed to model the water flow dispersal process to explain the macrophyte species community in a multi-pond system. To achieve this, we explored four different spatial eigenfunction approaches: Euclidean distance-based spatial eigenfunction, watercourse distance-based spatial eigenfunction, asymmetric eigenvector maps (AEM) and watercourse distance-based AEM (Fig. 2). Additionally, we generated the spatial eigenfunctions for the common versus rare species separately to assess which spatial patterns best explained the common and rare macrophyte community.

Watercourse connectivity variables were calculated using the AEM package in R (R Development Core Team, 2016) via AEM analyses as described by Blanchet et al. (2008). These spatial variables were derived from a matrix of hydrological connection diagram based on the presence or absence of links (i.e., ditches) between pairs of study ponds (Blanchet et al., 2008, 2011). Given that all ditches in the study area are used for agricultural irrigation, we assumed that all connecting links had similar water flow rates. Our analysis revealed a moderate correlation of 0.59 between the overland and watercourse distance matrices.

To assess the unique contributions of the four spatial eigenvectors generated by different spatial matrices, we performed three partial RDA models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◂**Fig. 2** Different ecological spatial process for the macrophyte dispersal among the hydrological-connected ponds. Four spatial eigenfunctions were employed to illustrate various ecological spatial process in this study. **Fig. 2a** The Euclidean distance-based MEM emphasizes non-directional dispersal via overland distance, which is useful for examining the movement of terrestrial organisms such as those transported by **wind or animals**. **Fig. 2b** The watercourse distance-based MEM represents the non-directional dispersal via watercourse distance, which is better suited for investigating the movement of aquatic organisms such as those transported by **flying insects or animals**. **Fig. 2c** The 0/1 based AEM emphasizes **flow direction** makes it a useful tool for analyzing the movement of aquatic species through channel networks or via water flow. **Fig. 2d** The weighted AEM based on watercourse distance emphasizes both flow direction and watercourse distance, providing insights into the movement of aquatic species through **channel networks** that are influenced by the **distance between watercourses** and **the rate of water flow**

using local environmental filtering data and each set of eigenvectors as independent variables. We applied these models to the total dataset, as well as separate analyses for the common and rare macrophyte species datasets. To compare the relative importance of environmental filtering and spatial processes for common and rare macrophyte species, we partitioned the variance between the environmental dataset and different spatial eigenvector datasets. We used variation partitioning to estimate the most significant determinants of macrophyte species relative to environmental, land use, hydrological connectivity and landscape connectivity variables (Table 1). The total variation of the macrophyte community was partitioned into the shared and pure contributions of the four explanatory variable datasets: local environment, land use, hydrological connectivity, and landscape connectivity variables. By examining these effects, researchers can gain insights into the underlying mechanisms that shape the occurrence and abundance of common and rare species, providing a more holistic understanding of macrophyte community dynamics. We conducted all analyses using R with the packages *packfor*, *spacemaker*, *ade4*, *vegan*, and *AEM* (Borcard et al., 2011).

Results

Contribution of common and rare species to richness pattern of macrophytes in multi-pond system

Figure 3 displayed the ranking abundance curve of the 62 macrophyte species from the most common to the rarest. The results demonstrated that the correlation

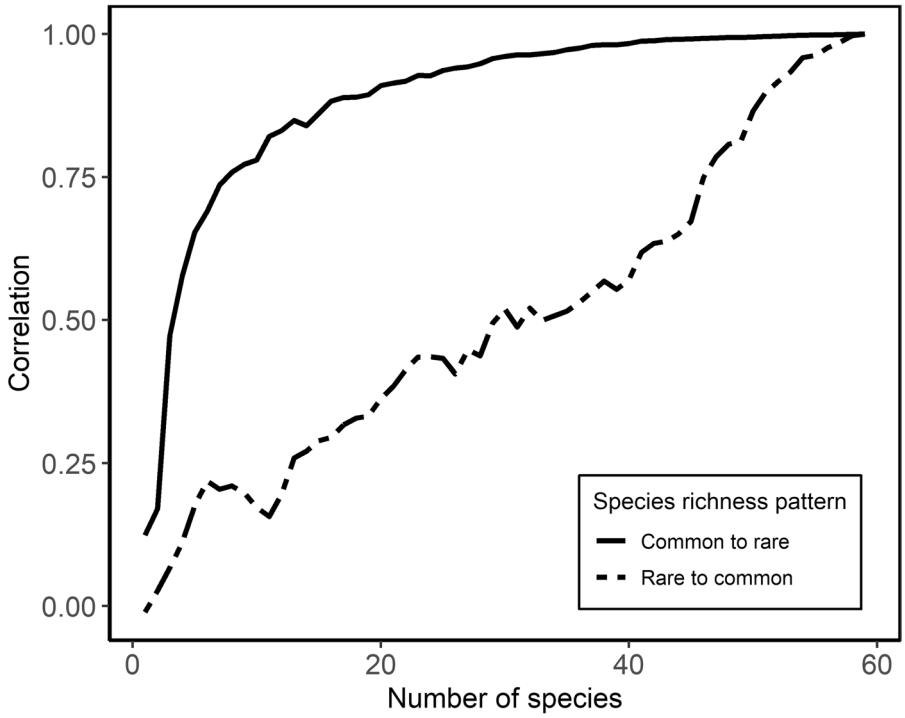

Fig. 3 The relative contribution of rare and common macrophyte species to overall species richness patterns in the multi-pond system. The sequential correlations were measured between the partial and full assemblage macrophyte species richness patterns. Sub-assemblage of macrophyte species was assembled by adding one species at a time and was correlated with the full assemblage of macrophyte species. The species was added following the order from common-to-rare and from rare-to-common in the figure

between the sub-assemblage and overall macrophyte species richness increased more rapidly for the common-to-rare sequences than for the rare-to-common sequences (Fig. 3). Although the increasing rate of the rare-to-common sequence was slower than that of the common-to-rare sequence, the effect of rare species was relatively apparent in the rare-to-common sequence, indicating that common species provided a better approximation to overall species richness than rare species. Notably, a few slices early along the rare-to-common sequences showed negative correlations with the overall richness patterns, while the common-to-rare sequences exhibited stable increases

Effect of connectivity and environmental factors on the composition of common and rare macrophyte species

Table 3 indicated that the watercourse distance-based AEM model (adj-$r^2 = 0.0512$) could better explain the variance of macrophyte communities compared to both the 0/1 AEM model (adj-$r^2 = 0.0321$) and the watercourse distance-based MEM model (adj-$r^2 = 0.0083$). These results suggest that both the direction of water flow and water course distance play a more critical role in the dispersal of macrophyte communities in the multi-pond system than dispersal by wind or flying animals. The simulation procedure was effective in identifying the optimal spatial eigenvector for modeling macrophyte community dispersal with water flow.

For the total macrophyte species, there was a considerable difference between the adj-r^2 value obtained with the environmental, land use and spatial variables generated using the Euclidean and hydrological distance (see Figure S2 for more information). Of these, the environmental dataset was found to be a vital factor in explaining the variance in macrophyte species community (3.75%). The spatial variables with the spatial eigenvector generated from the watercourse distance-based AEM explained 5.12% of the total variance, while the spatial eigenvector based on Euclidean distance explained only 2.19% of the variance in macrophyte community (Figure S3).

When partitioning the variance of the common and rare macrophyte species datasets, different patterns emerged. The adjusted R-square values of the rare species models were consistently lower than those of the common species models. In the case of common species, the species variation was driven mainly by water flow weighted by the watercourse distance (5.19%), in combination with the environmental variables (5.00%). Conversely, for rare species, the models based on Euclidean distance retained more spatial variables (3.01%). Surprisingly, the water connectivity did not play an important role in explaining the variance of the rare species (1.59%). Variation partitioning of the Hellinger-transformed macrophyte data into an environmental component, a land-use component, a hydrological component, and a spatial component is shown in Figure S2 in supporting information.

Discussion

This study has the potential to deepen our understanding of how hydrological and landscape connectivity influences the structure of macrophyte species in ponds and seeks to shed light on the mechanisms through which connectivity contributes to maintain regional biodiversity (Akasaka et al., 2010). Hydrological connectivity likely plays a pivotal role in explaining species turnover between ponds that share direct hydrologic connections (Miller et al., 2002; Short & Caterino, 2009; Murphy et al., 2010; Chester et al., 2015; Mims et al., 2015). However, when two lakes are geographically proximate yet linked by a lengthy stretch of river channel, the importance of hydrological distance may be overshadowed by that of geographical distance in determining species turnover. This study employed variation partitioning to compare different dispersal modes in elucidating macrophyte community structure, with a particular focus on common versus rare species. This study used variation partitioning to examine how different dispersal modes shape macrophyte community structure, focusing on the contrasting responses of common and rare species. By highlighting the interplay between connectivity and dispersal, our findings advanced understanding of how river networks facilitate species movement, enabling both common and rare species to access diverse habitats (Datry et al., 2016). The pivotal role of hydrological connectivity in structuring macrophyte communities supported metacommunity theory, demonstrating the influence of spatial processes even in human-modified landscapes. Moreover, the distinct responses of common and

Table 3 Results of variation partitioning analyses represented on environmental and spatial models for explaining the total macrophyte, common species and rare species community

Data	Hydrological connectivity types	Hydrological connectivity variables retained	Landscape connectivity variables retained (Euclidean distance-based MEM)	Environmental variables retained	Land use	Fraction (adj-r^2)			
						Envi	Land use	Hydrological connectivity	Landscape connectivity
Total	Watercourse distance-based MEM	3, Inflow, Outflow, WRT	MEM3, MEM2, MEM12, MEM9, MEM7, PC, IIC	DO, Elevation, pH, ORPmv, Transparency	Farmland %, Building%	0.0555	0.0266	0.0083	0.0304
	0/1 AEM	1, 2, 5, 4, 20, 65, 25, 106, Inflow, Outflow, WRT				0.0307	0.0089	0.0321	0.0119
	Watercourse distance-based AEM	2, 21, 7, 1, 6, 66, 16, Inflow, Outflow, WRT				**0.0375**	0.0198	**0.0512**	0.0219
Rare species	Watercourse distance-based MEM	2, Inflow, Outflow, WRT	MEM2, MEM7, MEM4, MEM3, MEM11, PC, IIC	Temperature, Elevation, ORPmV	Farmland %, Building%	0.0140	0.0027	0.0061	0.0345
	0/1 AEM	85, 1, 4, Inflow, Outflow, WRT				0.0182	0.0056	0.0301	0.0332
	Watercourse distance-based AEM	58, 10, 47, 7, Inflow, Outflow, WRT				0.0154	***NS***	0.0159	**0.0301**
Common species	Watercourse distance-based MEM	3, Inflow, Outflow, WRT	MEM3, MEM2, MEM12, MEM9, PC, IIC	DO, Elevation, pH, ORPmV, Transparency	Farmland %, Building%	0.0721	0.0346	0.0056	0.0260
	0/1 AEM	2, 1, 20, 65, 25, 4, 54, 10, 13, Inflow, Outflow, WRT				0.0352	0.0112	0.0445	0.0247
	Watercourse distance-based AEM	2, 21, 1, 7, 6, 5, 25, 16, 47, 65, Inflow, Outflow, WRT				**0.0500**	0.0228	**0.0519**	0.0296

Two MEM models of the Euclidean distance-based MEM models and the watercourse distance-based MEM models and Two AEM models of the 0/1 AEM and the Watercourse distance-based AEM models were performed in analyses. We retained spatial variables that had high spatial eigenvalues. The spatial variables were represented in the order they were maintained in the stepwise procedure. The fraction of variance explained by each variable (i.e., Envi—environment, Land use—land use, Hydrological connectivity—Hydrological spatial eigenvector, and Landscape connectivity—overland spatial eigenvector) was measured using adj-r^2. Non-significant variables were marked as "NS," which means the factors are non-significant correlated with the macrophyte community

rare species to environmental and spatial drivers offered valuable insights into niche theory, underscoring the importance of considering both dispersal limitations and environmental filtering in biodiversity management.

Species richness pattern of macrophyte in the multi-pond system

Our results showed that the macrophytes exhibited a common-to-rare sequence that was more strongly correlated with total species richness patterns than the rare-to-common sequence, thus corroborating our hypothesis (1) that common macrophyte species provide a better approximation to overall species richness than the rare species. This finding was consistent with previous studies which emphasized the importance of common species in shaping patterns of species richness at both smaller and coarser scales (Lennon et al., 2004; Landi & Chiarucci, 2014; van Schalkwyk et al., 2019). Common species were often used as indicators of overall species richness patterns and should be prioritized for protection (Pearman & Weber, 2007; Landi & Chiarucci, 2014). Common species adapt to the prevailing environmental conditions and are resilient to disturbances, making them more likely to provide consistent ecosystem services such as oxygen production, carbon sequestration, and sediment stabilization, and support the stability of the overall freshwater ecosystem (Van Schalkwyk et al., 2019). Common macrophyte species always act as indicators of environmental changes or pollution in the aquatic ecosystem, and monitoring common species helps identify ecosystem stressors promptly, preventing further degradation (Bucior et al., 2021).

In contrast, rare species occur in only a few specific quadrats, and their loss would be limited to those individual quadrats (Pérez-Quesada & Brazeiro, 2013). Nonetheless, the importance of rare species in shaping macrophyte community diversity patterns could not be ignored. Previous studies have highlighted the ability of rare species to replace dominant species after disturbances and their ecological functions in the ecosystem (Chapman et al., 2018). Rare macrophyte species have unique ecological roles and contribute to biodiversity; their limited distribution and abundance mean that their influence on the total species richness patterns and ecosystem-level processes may be more localized and niche-specific. Rare species have specialized adaptations with specific organisms and play unique ecological roles in niche habitats that common species cannot replace (Säterberg et al., 2019). Rare species also harbor genetic traits, which is essential for the long-term adaptation and survival of species in response to environmental changes (Dee et al., 2019).

Conservation efforts aimed at preserving both common and rare macrophyte species are essential for maintaining overall biodiversity and ecosystem resilience. However, prioritizing the preservation and restoration of common species ensures that the fundamental ecosystem functions and services they provide are safeguarded for the benefit of a broader range of human communities.

Role of environmental process for common and rare species

Earlier studies have demonstrated that both common and rare macrophyte species are predominantly influenced by local environmental factors at a fine-grained scale (Siqueira et al., 2012). Our study further revealed that environmental factors played a more significant role in shaping common macrophyte species compared to rare species, supporting Hypothesis (2). This could be explained by the 'Niche-width hypothesis,' which suggests common species has a wide ecological niche and could tolerate a wide range of environmental conditions and make use various localized resources (Vázquez & Gaston, 2004, Polechová & Barton, 2015). Common macrophyte species typically have short generation times, high reproductive output, and effective dispersal mechanisms, which allow them to dominate and outcompete other species in suitable habitats (Ahmad et al., 2021). The high abundance and widespread distribution of the common species make them contributing significantly to ecological processes like nutrient cycling, habitat provision, and trophic interactions in aquatic ecosystem. It is worth noting that environmental variables also influenced rare macrophyte species from our results. The limited distribution and lower abundance of rare species make it more challenging to detect clear patterns or relationships with environmental factors. Additionally,

factors like genetic isolation and stochastic events can have a more pronounced impact on rare species than direct environmental influences (Ahmad et al., 2021).

The macrophyte species composition is influenced by a complex interplay of environmental factors (Sun et al., 2019). Among different environmental variables, our results indicated that elevation was a significant factor in structuring the patterns of both common and rare macrophyte species in the multi-pond system. Elevation acts as an environmental gradient that influences various ecological factors in aquatic ecosystems (Sun et al., 2022). These factors collectively determine the distribution and abundance of macrophyte species at different elevations, resulting in distinct species compositions along elevation gradients. For example, the elevation gradient creates distinct temperature and light environment, leading to the establishment of macrophyte species with varying light requirements for photosynthesis at different elevations (Herb & Stefan, 2003).

Role of connectivity for common and rare species

Pond connectivity facilitates species dispersal and community assembly by enabling the movement of organisms and propagules among ponds (Ribeiro et al., 2011; Jahanishakib et al., 2021). While common macrophyte species often benefit from frequent dispersal via multiple vectors such as waterbirds, wind, and surface flow, rare species may be more constrained by limited vector availability and reduced connectivity (Morris, 2012). To better understand these dynamics, we applied four spatial eigenfunction approaches to model pond connectivity and simulate macrophyte dispersal. Our results revealed that the relative importance of spatial effects varies between common and rare macrophyte species, possibly due to differences in their spatial range and dispersal abilities across taxonomic groups.

In line with hypothesis (3), our results indicated that watercourse distance-based AEM variables were generally more influential in explaining the composition of common species than rare species, indicating both the direction of water flow and watercourse distance are vital in structuring the common species. The result was supported by previous study reported common macrophyte species typically produce propagules that have excellent flotation capabilities, enabling them to disperse over long distances by water flow (Dahlgren & Ehrlen, 2005). Specifically, the direction of water flow plays a critical role in the dispersal of macrophyte propagules (seeds, spores, or vegetative fragments) for common species (Dawson, 1988). Seeds or propagules released by macrophytes in upstream areas can be carried downstream by the water flow, allowing for colonization of new habitat (Chen et al., 2007). Closer watercourse distance between ponds allows for easier movement of macrophyte propagules, enhancing gene flow and colonization (Cornacchia et al., 2019). The direction of water flow and watercourse distance play a crucial role in shaping the community structure and overall biodiversity of macrophytes in the multi-pond system.

We also found the Euclidean distance-based spatial variables were relatively more important in explaining the species composition for rare species, suggesting that rare species are more rely on the wind and water bird for dispersal between the ponds. Unlike common species tend to occupy a greater number of habitats and maintain higher abundances (Brown, 1984), rare species demonstrate a higher level of environmental favorability, they may become abundant in specific sites (Brown & Kodric-Brown, 1977; Mouillot et al., 2021). They produce seeds or propagules that are not well-suited for dispersal by water currents, limiting their ability to spread to new habitats. Rare macrophyte species may have specialized mechanisms for dispersal on their own, and they rely on external agents like wind and water birds. Wind and water birds can transport propagules over distances and facilitate gene flow between isolated populations. Macrophytes that produce lightweight seeds or spores can take advantage of wind currents to disperse their propagules to neighboring ponds or even distant ponds (Hovenden et al., 2008). Water birds are highly mobile and often cover large distances during migration and foraging activities (Clausen et al., 2002). Water animals could pick up macrophyte propagules on their feathers, feet, or beaks and inadvertently transport them to new locations. This unintentional transport can lead to the establishment of rare macrophyte populations in isolated habitats. Overall, rare species mainly rely on wind and water birds for dispersal between ponds due to their specific adaptations, and the opportunities presented by the mobility of these dispersal agents. Understanding these dispersal mechanisms is crucial for conservation efforts aimed at preserving and restoring rare macrophyte populations and promoting their long-term survival. Our study acknowledges limitations, including low adj-r^2

values, which reflect unmeasured factors and stochastic processes influencing macrophyte community structure. While significant partitioned fractions confirm non-random patterns, a larger dataset across diverse landscapes could improve the predictive accuracy, particularly for rare species. Future study would also explore alternative dispersal mechanisms, such as animal- or wind-mediated dispersal, to enhance understanding of community assembly and inform conservation strategies. Another promising direction involves investigating how landscape configurations affect macrophyte biodiversity under land-use changes and climate variability. Moreover, understanding how rare species respond to shifts in connectivity and environmental gradients remains a key research gap that could further refine biodiversity management in multi-pond systems.

Conclusion

This study investigated the species richness patterns and factors shaping the species composition of macrophyte communities in an agricultural multi-pond system. Environmental factors, particularly elevation, were found to be important in shaping macrophyte species composition. The relative importance of spatial effects varied between common and rare species. Watercourse-dispersal processes are strongly associated with the common species, while Euclidean distance (i.e., dispersal by animals or wind) is a significant factor in explaining rare species communities. Our findings highlighted key management strategies for conserving macrophyte biodiversity in fragmented aquatic systems. Maintaining or restoring hydrological connectivity between ponds was essential to facilitate species dispersal, promoting biodiversity resilience in multi-pond landscapes. Additionally, managing water quality by controlling transparency and nutrient levels could help sustain species diversity, particularly for rare species vulnerable to eutrophication. Prioritizing pond conservation based on connectivity and environmental conditions could guide targeted restoration efforts and land-use practices, balancing biodiversity conservation with agricultural productivity.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval Not applicable.

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